Arista Access Node Failure SOP: BAS-ACC-10 at Basavanagudi POP for Optical Transport Failure (ALARM_1093)

Node: BAS-ACC-10 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_1093

Description

This procedure outlines the steps to diagnose and resolve optical transport failures on the Arista Access Node, BAS-ACC-10, located at the Basavanagudi POP. The failure domain in question is characterized by the degradation of SFP/XFP interfaces, fiber cuts, or loss of Rx/Tx power. This failure domain is specific to the Access role on Arista equipment due to the reliance on optical transport as a critical component of data transmission. The underlying protocol mechanics involve the use of Layer 1 protocols, such as SONET/SDH, to provide high-speed data transmission over optical fibers. The degradation or failure of these interfaces can have a cascading effect on the overall network, resulting in packet loss, increased latency, and reduced network availability. Therefore, prompt diagnosis and resolution of this failure domain are crucial to maintaining network uptime and ensuring the delivery of critical services.

Pre-requisite Checks

- Check: Verify that remote hands have been dispatched to the Basavanagudi POP to assist with physical interventions.

- Check: Check the status of redundant links to ensure that they are operational and can take over in the event of a failure.

- Check: Verify that all necessary permissions and access have been granted to perform the necessary diagnostic and resolution steps.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces detail, Expected Output: Interface
information, including SFP/XFP status and optical power levels,
Description: This command displays detailed information about the
interfaces on the Access Node, including the status of SFP/XFP
interfaces and optical power levels. This information can help
identify potential issues with the optical transport.
```

```
Command: show fiber status, Expected Output: Fiber status, including cuts and faults, Description: This command displays the status of the fibers connected to the Access Node, including any cuts or faults that may be affecting the optical transport.

Command: show optics, Expected Output: Optics information, including temperature and voltage, Description: This command displays information about the optics used in the SFP/XFP interfaces, including temperature and voltage levels. This information can help identify potential issues with the optical transport.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP interfaces, Command: replace interface , Description: This step involves replacing the SFP/XFP interfaces on the Access Node to restore optical transport functionality. The remote hands will need to physically access the Access Node to perform this replacement.

Step 2: Step: Repair or replace fiber, Command: repair fiber , Description: This step involves repairing or replacing the fiber that has been cut or damaged. The remote hands will need to physically access the affected area to perform this repair or replacement.

Step 3: Step: Adjust Rx/Tx power levels, Command: set optics rx-power tx-power , Description: This step involves adjusting the Rx/Tx power levels on the SFP/XFP interfaces to restore optimal optical transport functionality.

Post-Check Verification

```
Command: show interfaces detail, Expected Output: Interface information, including SFP/XFP status and optical power levels, Description: This command is used to verify that the SFP/XFP interfaces are functioning correctly and that the optical power levels are within the optimal range.

Command: show fiber status, Expected Output: Fiber status, including cuts and faults, Description: This command is used to verify that the fibers are functioning correctly and that there are no cuts or faults affecting the optical transport.

Command: show optics, Expected Output: Optics information, including temperature and voltage, Description: This command is used to verify that the optics used in the SFP/XFP interfaces are functioning correctly and that the temperature and voltage levels are within the optimal range.
```

Roll-back Procedures

- Step: Restore original SFP/XFP interfaces, Command: undo replace interface , Description: This step involves restoring the original SFP/XFP interfaces on the Access Node in the event that the replacement interfaces do not resolve the issue.

- Step: Restore original fiber, Command: undo repair fiber , Description: This step involves restoring the original fiber in the event that the repair or replacement fiber does not resolve the issue.

- Step: Restore original Rx/Tx power levels, Command: undo set optics rx-power tx-power , Description: This step involves restoring the original Rx/Tx power levels on the SFP/XFP interfaces in the event that the adjustment does not resolve the issue.